

1B Study Guide: What Is Life? Metabolism, Homeostasis, and Feedback

Learning target 1B

(Textbook 1.3) Scientific investigation includes a variety of methods to develop explanations that are based on evidence and revised through argumentation.

How to use this guide

You don't need the textbook to use this guide. Every textbook page that matters for 1B is copied into **Part 1**, with the important sentences highlighted. Each highlight has a numbered badge that matches a note in the column beside the page. The notes explain hard words, answer the textbook's questions, and connect the page to what your teacher asks about in class.

Part 1 has three groups of pages:

- **Lesson 1.3** (pp. 22–25 and the review on p. 29): what makes something alive, metabolism, homeostasis. This is the lesson the unit guide lists for 1B.
- **Chapter 27** (pp. 907, 908, 932): how feedback keeps the body stable. The class textbook explains feedback here, much later in the book.
- **Lessons 1.1 and 1.2** (pp. 12 and 21): controlled experiments, models, and arguing from evidence, the “scientific investigation” half of the 1B target.

After Part 1 come the other sources (Campbell covers positive feedback, which the class textbook doesn't), a summary, the vocabulary explained in detail, and full answers to the teacher's study questions. Cover the answers in Part 5 and try each question out loud before you check.

Notes are color-coded:

1 Key idea **2** Vocabulary **3** Make it make sense **4** Watch out **5** Connection

What this guide covers

Where to read

Source	Section	Pages	What it covers
Class textbook: <i>Miller & Levine Biology</i> (2019)	Lesson 1.3 Patterns of Life (the lesson the unit guide lists)	22–25	characteristics of living things, metabolism, homeostasis, viruses, stability and change, mechanism, system models
Class textbook	Lesson 1.1 What Is Science?	12	controlled experiments, control groups

Source	Section	Pages	What it covers
Class textbook	Lesson 1.2 Science in Context	21	models, argument from evidence
Class textbook	Lesson 27.1 Organization of the Human Body	907–908	homeostasis, negative feedback (feedback inhibition), thermostat, body temperature
Class textbook	Lesson 27.3 Human Systems II	932	negative feedback in water balance and blood glucose, feedback loop
Reference: <i>Campbell Biology</i>	Ch. 1, Concept 1.1	3, 9	properties of life, negative and positive feedback
Reference: <i>Campbell Biology</i>	Ch. 8, Concept 8.1	146	metabolism, catabolic and anabolic pathways
Reference: <i>Campbell Biology</i>	Ch. 40, Concept 40.2	893–894	homeostasis, set point, dynamic equilibrium, positive feedback in childbirth
Teacher materials	What Is Life? video lecture and skeleton notes	—	characteristics of living things, viruses
Teacher materials	Feedback Mechanisms and Homeostasis slides; Feedback practice	slides 2–10	feedback loops, body temperature, childbirth, climate feedbacks

Lesson 1.3 introduces homeostasis in one paragraph but doesn't explain feedback. Feedback loops are covered much later in the class textbook (Chapter 27, the human body), and **positive feedback isn't covered in Miller & Levine at all**. Those parts use Chapter 27, Campbell, and the teacher's slides.

Lesson 1.3 Patterns of Life

Miller & Levine Biology, textbook page 22

The lesson the unit guide lists for 1B. This page answers the first study question: what are the criteria for something to be alive?

1 LESSON VOCABULARY

For 1B, the key words are **metabolism**, **stimulus**, and **homeostasis**. Each one is explained on p. 24.

2 KEY IDEA

The big question of this lesson: what makes something **alive**? It's harder to answer than it sounds.

3 WATCH OUT

You can't use **one** trait to decide if something is alive. Living things show **all** of the characteristics together.

4 MAKE IT MAKE SENSE

A fire gives off light, moves, "grows," and uses energy, but it isn't alive: it has no cells, no DNA, and no homeostasis. That's why you need the whole list.

5 CONNECTION

Viruses are the classic borderline case. The What Is Life? lecture asks whether they're alive. See p. 24 and the answer to study question 1.

6 THE 8 CRITERIA FOR LIFE

1 made of cells, 2 reproduce, 3 universal genetic code (DNA), 4 grow and develop, 5 obtain and use materials and energy, 6 respond to the environment, 7 keep a stable internal environment, 8 evolve. Memorize this list; it answers study question 1.

LESSON 1.3 Patterns of Life

KEY QUESTIONS

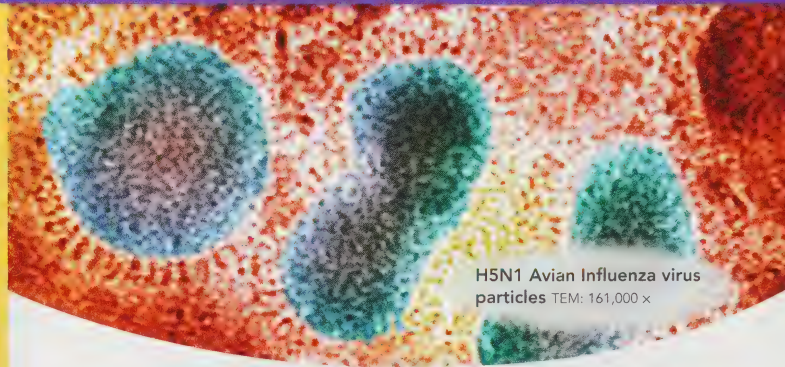
- What characteristics do all living things share?
- What are the crosscutting concepts of biology?
- How do fields of biology differ in their approaches?
- How is the metric system important to science?

VOCABULARY

- biology
- sexual reproduction
- asexual reproduction
- DNA
- metabolism
- stimulus
- homeostasis
- evolve

READING TOOL

Complete the chart in your **Biology Foundations Workbook** by writing the main idea and details from this lesson on the characteristics of life.



Think about some of the important and exciting news stories you've read or heard about recently. Bird flu spreads around the world, killing thousands of birds and threatening a human epidemic. Users of certain illegal drugs experience permanent damage to their brains and other parts of their nervous systems. Reports surface about the use of cloned human cells to grow new organs to replace those lost to disease or injury. These and many other stories involve **biology**—the science that employs scientific methodology to study living things. (The Greek word *bios* means "life," and *-logy* means "study of.")

Characteristics of Living Things

- Biology is the study of life. But what is life? What distinguishes living things from nonliving matter? Surprisingly, it isn't as simple as you might think to describe what makes something alive. **No single characteristic is enough to describe a living thing.** Also, some nonliving things share one or more traits with organisms. For example, **a firefly and a fire both give off light**, and each moves in its own way. Mechanical toys, automobiles, and clouds (none of which are alive) move around, while mushrooms and trees (which are alive) stay in one spot. To make matters more complicated, some things, such as **viruses**, exist at the border between living and nonliving things.

- Despite these difficulties, we can list characteristics that most living things have in common. **Living things are made up of basic units called cells, reproduce, are based on a universal genetic code, grow and develop, obtain and use materials and energy, respond to their environment, maintain a stable internal environment, and evolve, changing over time.**

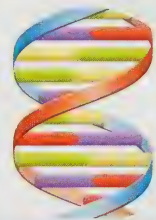
7 Made Up of Cells Living things, or organisms, are made up of small, self-contained units called cells. Cells are the smallest units of an organism that can be considered alive. Cells can grow, respond to their surroundings, and reproduce. Despite their small size, cells are complex and highly organized.

Reproduction **8** All organisms produce new organisms through a process called reproduction. There are two basic kinds of reproduction: sexual and asexual. The vast majority of multicellular organisms reproduce sexually. In **sexual reproduction**, cells from two different parents unite to produce the first cell of the new organism. In **asexual reproduction**, the new organism has a single parent.

Based on a Universal Genetic

Code Offspring usually resemble their parents. With asexual reproduction, offspring and their parents have the same traits. With sexual reproduction, offspring differ from their parents in some ways. Explaining how organisms inherit traits is one of the greatest

9 achievements of modern biology. Biologists now know that the directions for inheritance are carried by a molecule called **DNA**. This genetic code, with a few minor variations, determines the inherited traits of every organism on Earth.



DNA helix

10 Growth and Development All living things grow during at least part of their lives. For some single-celled organisms, such as bacteria, growth is mostly a simple increase in size. Multicellular organisms, like this monarch butterfly, typically go through a process called development. During development, a single fertilized egg cell divides again and again to produce the many cells of mature organisms. As those cells divide, they change in shape and structure. This process is called differentiation, because it forms cells that look different from one another and perform different functions.

VIDEO

Learn about the characteristics of life and what it means to be alive.

Lesson 1.3 Patterns of Life

Miller & Levine Biology, textbook page 23

The first four characteristics of life.

7 ORGANISM · CELLS (CRITERION 1)

An **organism** is an individual living thing. The **cell** is the smallest unit that is alive. Unicellular organisms are one cell (bacteria); multicellular ones have many (you).

8 REPRODUCTION (CRITERION 2)

Sexual: two parents; offspring differ from them. **Asexual:** one parent; offspring are copies. A sterile mule can't reproduce but is still alive, because the criteria describe living things in general, not every individual.

9 GENETIC CODE (CRITERION 3)

Instructions for traits are stored in **DNA**, and nearly every organism uses the same code. That's why it's called *universal*.

10 GROW AND DEVELOP (CRITERION 4)

Growth is getting bigger. **Development** is changing form: one fertilized egg divides into many cells that **differentiate** into different types (egg → caterpillar → butterfly).





Need for Materials and Energy

Think of what an organism needs as it grows and develops. Just as a building grows taller because workers use energy to assemble new materials, an organism uses energy and a constantly supply of materials to grow, develop, and reproduce. Organisms also need materials and energy just to stay alive. The combination of chemical reactions as it carries out its life processes is called **metabolism**.

Response to the Environment

Organisms detect and respond to stimuli from their environment. A **stimulus** is a signal to which an organism responds. External stimuli, which come from the environment outside an organism, include factors such as light and temperature. In contrast, internal stimuli come from within an organism. This field of sunflowers is responding to the stimulus of sunlight.

Maintaining Internal Balance Even though conditions in the external environment may vary widely, most organisms must keep internal conditions, such as temperature and water content, fairly constant to survive. The relatively constant set of conditions is called **homeostasis**. Often internal stimuli help maintain homeostasis. For example, when your body needs more water to maintain homeostasis, internal stimuli make you feel thirsty.

Evolution Although individual organisms experience many changes during their lives, the basic traits they inherited from their parents usually do not change. As a group, however, any given kind of organism can **evolve**, or change over time.

Over a few generations, the changes in a group may not seem significant. But over hundreds of thousands or even millions of years, the changes can be dramatic. The ability of a group of organisms to change over time is invaluable for survival in a world that is always changing.

What About Viruses? Viruses are particles made up of proteins, nucleic acids, and sometimes lipids. Viruses depend entirely upon other living organisms for their existence, making them parasites. After infecting living things, viruses can reproduce, regulate gene expression, and evolve.

READING CHECK **16 Explain** Why is it important for organisms to maintain homeostasis?



Lesson 1.3 Patterns of Life

Miller & Levine Biology, textbook page 24

The last four characteristics, and the definitions of metabolism, stimulus, and homeostasis.

11 METABOLISM (CRITERION 5)

Metabolism is *all* the chemical reactions in an organism, both building molecules up (photosynthesis, protein synthesis) and breaking them down (cellular respiration, digestion). This is study question 2.

12 STIMULUS (CRITERION 6)

A **stimulus** is a signal an organism reacts to. **External:** light, temperature (sunflowers turning toward the sun). **Internal:** from inside the body (thirst). The plural is *stimuli*.

13 HOMEOSTASIS (CRITERION 7)

Homeostasis means keeping internal conditions (temperature, water) *fairly constant* even when the outside changes. Thirst is an internal stimulus that tells you to drink and restore water balance. It's a **feedback loop**, explained on pp. 907–908.

14 EVOLUTION (CRITERION 8)

Groups evolve over many generations; **individuals** do not. Your traits don't evolve during your lifetime.

15 ARE VIRUSES ALIVE?

Viruses **reproduce and evolve**, but only inside a host cell. They have **no cells and no metabolism** of their own. Most biologists call them nonliving, or "at the border." Pick a side and support it with the criteria (CER).

16 READING CHECK: ANSWER

Cells only work within a narrow range of conditions. If temperature or water levels go too far out of range, cells can't carry out metabolism and the organism can get sick or die (for example, heat-stroke or dehydration).

The units of this book seem to cover different subjects, but all biological sciences are tied together by themes and methods of study that cut across disciplines. These concepts overlap and interlock, and crop up throughout the book. You'll also notice that several of these concepts overlap with the characteristics of life or the nature of science.

18 Cause and Effect: Mechanism and Explanation Science is not a list of facts, but “a way of knowing.” The job of science is to use observations, questions, and experiments to explain the natural world in terms of natural forces and events, or cause and effect. Successful scientific research reveals rules and patterns that can explain and predict at least some events in nature. Science enables us to take actions that affect events in the world around us.

Stability and Change ²⁰ Living things maintain a relatively stable internal environment, a process known as homeostasis. For most organisms, any breakdown of homeostasis may have serious or even fatal consequences.

Scale, Proportion, and Quantity Just about everything, from atoms to rocks to organisms, can be described in terms of structure. How something's structure is studied and described depends on the scale on which it is being studied. On a molecular scale, a scientist might study a human in terms of proteins in the body. On a global scale, a scientist might study humans in terms of their impact on the ozone layer.

Throughout nature you will find patterns. Similar patterns can be in structures ranging from an unrolling fern frond to a shell to a pine cone.



Miller & Levine Biology, textbook page 25

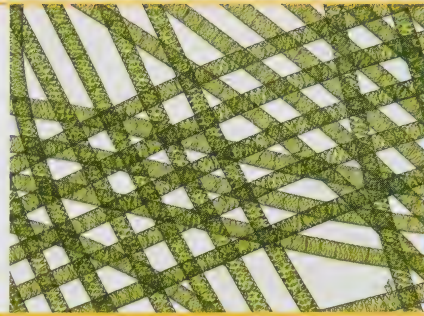
Homeostasis keeps the body **stable** while conditions **change**. If homeostasis breaks down, the result can be serious or fatal, which is why feedback matters (study question 3).

**Develop a Solution Lab****Open-ended Inquiry****Algae in the Water**

Problem What are the ideal growth conditions for algae?

In this lab, you will plan and conduct an experiment to determine the effects of a variable, such as temperature or sunlight, on algae growth. Then you will propose ideas for a system for raising algae.

You can find this lab in your digital course.



Safety Scientists, such as those shown in **Figure 1-14**, are trained to use safety procedures, such as wearing protective glasses and gloves. Laboratory work may involve flames or heating elements, electricity, chemicals, hot liquids, sharp instruments, or breakable glassware. Laboratory work and fieldwork may involve contact with living or dead organisms—not just potentially poisonous plants and venomous animals but also disease-carrying organisms and substances contaminated with dangerous microorganisms.

Whenever you work in your biology laboratory, you must follow safe practices. Careful preparation is the key to staying safe. Before performing any activity in this course, study the safety rules in the Lab Skills Handbook. Before you start each activity, read all the steps and make sure you understand them, including safety precautions.

The single most important safety rule is to always follow your teacher's instructions and directions in this textbook. Anytime you are in doubt, ask your teacher for an explanation. And because you may come in contact with organisms you cannot see, it is essential that you wash your hands thoroughly with soap and warm water after every scientific activity. Remember that you are responsible for your own safety and that of your teacher and classmates. If you are handling live animals, you are responsible for their safety, too.

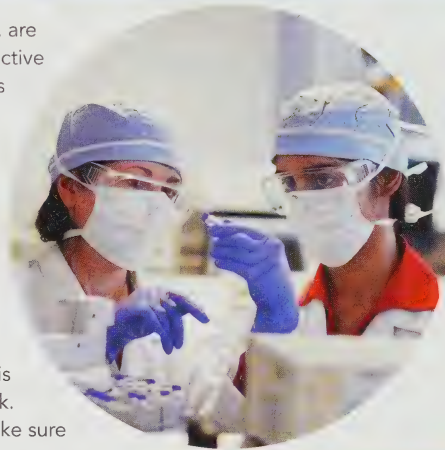
**INTERACTIVITY**

Figure 1-14

Safety in the Laboratory

These scientists are wearing protective glasses and gloves in keeping with the safety protocols in their lab.

Lesson 1.3 Patterns of Life

Miller & Levine Biology, textbook page 29

The end of Lesson 1.3. The top half is lab safety (not on the study questions). The notes answer the lesson review.

21 REVIEW 1–3: ANSWERS

1. Living things have *all* the characteristics of life: cells, reproduction, DNA, growth and development, use of materials and energy, response to stimuli, homeostasis, and evolution. Nonliving things may share a few but never all. **2.** The crosscutting concepts connect all branches of biology with the same big ideas (cause and effect, systems, stability and change, ...). **3.** Biology can be studied at every scale, from molecules and cells to organisms, ecosystems, and the whole planet.

22 REVIEW 4–6: ANSWERS

4. So scientists everywhere can compare and **replicate** each other's measurements (the metric system). **5.** All organisms share the same molecules and cell processes, so patterns at that level explain features of all life. **6.** $250 \text{ g} \times 10 = 2,500 \text{ g} = 2.5 \text{ kg}$.

**LESSON 1.3 Review****KEY QUESTIONS**

- 21** **1.** How do living things differ from nonliving things?
- 2.** How do crosscutting concepts help unite the study of biology?
- 3.** How can biology be studied at different scales, or levels?
- 22** **4.** Why do scientists use a common system of measurement?

CRITICAL THINKING

- 5. Identify Patterns** To analyze the characteristics of life, why is it useful to identify patterns at the level of molecules and cells?
- 6. Calculate** In an experiment, you need 250 grams of potting soil for each of 10 plant samples. How many kilograms of soil in total do you need?

Homeostasis

Some things about human activity are easy to observe. When you run or swim or write the answer to a test question, you can see your body at work. But behind the scenes, your body works constantly to do something that is difficult to see—maintaining a controlled,

- 23** stable internal environment. This process is called **homeostasis**, which means “keeping things the same.” *Homeostasis describes the relatively constant internal conditions that organisms maintain despite changes in internal and external environments.* Homeostasis may not be obvious, but for a living organism, it’s a matter of life or death.

- 25** **Feedback Inhibition** If you’ve ever watched someone driving a car down a straight road, you may have noticed the person slightly moving the wheel left and right, adjusting direction to keep the vehicle in the middle of the lane. In a sense, that’s how the systems of the body work, keeping internal conditions within a certain range, never allowing them to go too far one way or the other.

A Nonliving Example One way to understand homeostasis is to look at a nonliving system that keeps conditions within a certain range, such as a home heating system. In many homes, heat is supplied by a furnace.

- 26** When the temperature within the house drops below a set point, a thermostat sensor switches the furnace on. Heat produced by the furnace warms the house. When the temperature rises above the set point, the thermostat switches the furnace off, keeping the temperature within a narrow range.

- A system like this example is controlled by feedback inhibition. **Feedback inhibition**, or negative feedback, is the process in which a stimulus produces a response that opposes the original stimulus. **Figure 27-3** summarizes the feedback inhibition process in a home heating system. When the furnace is switched on, it produces a product (heat) that changes the environment of the house by raising the air temperature. This environmental change then “feeds back” to “inhibit” the operation of the furnace. In other words, heat from the furnace eventually raises the temperature high enough to trigger a feedback signal that
- 28** switches the furnace off. Systems controlled by feedback inhibition are generally very stable.

READING CHECK Apply Concept **29** Describe another example of a nonliving system that requires constant adjustment.

BUILD VOCABULARY

- 24** **Root Words** The root word *stasis* means “a state of balance or equilibrium.”

VIDEO

Learn how feedback inhibition is used to return the body to its normal, homeostatic state.

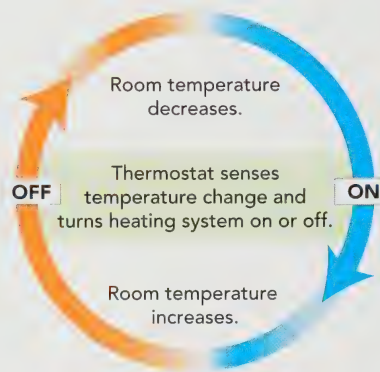


Figure 27-3
Feedback Inhibition

A home heating system uses a feedback loop to maintain a stable, comfortable environment within a house.

Lesson 27.1 Organization of the Human Body

Miller & Levine Biology, textbook page 907

From much later in the textbook: how homeostasis actually works. The key idea is negative feedback.

23 HOMEOSTASIS

Homeostasis = “keeping things the **same**”: relatively constant internal conditions, even when the environment inside or outside changes.

24 EQUILIBRIUM

Stasis means “a state of balance or **equilibrium**.” Homeo- (same) + stasis (balance) = staying in balance.

25 DYNAMIC EQUILIBRIUM

The car stays in its lane only because the driver keeps making small corrections. That’s **dynamic equilibrium**: stable overall, but always adjusting. The body keeps conditions within a **range** the same way.

26 MODEL · SET POINT

The thermostat is a **model** of homeostasis. The **set point** is the target temperature, and the **sensor** detects when the room drifts too far from it.

27 NEGATIVE FEEDBACK

Negative feedback (feedback inhibition): the response **opposes** the change. Too cold → heat on → warmer → heat off. It *negates* the change.

28 KEY IDEA

Negative feedback makes a system **stable**. That’s why organisms use it for homeostasis.

29 READING CHECK: ANSWER

Examples: a car’s cruise control (slows → more gas → speeds up → less gas), an air conditioner, an oven, or a toilet tank that refills until the float shuts the valve.

Lesson 27.1 Organization of the Human Body

Miller & Levine Biology, textbook page 908

The body's thermostat: negative feedback keeps your temperature near 37°C. This is the teacher's slide 4 example.



INTERACTIVITY

Figure 27-4

Body Temperature Control

In the human body, temperature is controlled through various feedback inhibition mechanisms.

30 A Living Example Could biological systems achieve homeostasis through feedback inhibition? Absolutely. One example is the maintenance of body temperature. **Figure 27-4** shows how the body regulates temperature in a way similar to that of a home heating system.

31 A part of the brain called the hypothalamus contains nerve cells that monitor body temperature. If the nerve cells sense that the core temperature has dropped much below 37°C, the hypothalamus produces chemicals that signal cells throughout the body to speed up their activities. Heat produced by this increase in activity, especially cellular respiration, causes a rise in body temperature, which is detected by nerve cells in the hypothalamus.

32 Have you ever been so cold that you began to shiver? If your body temperature drops well below its normal range, the hypothalamus releases chemicals that signal muscles just below the surface of the skin to contract involuntarily—to “shiver.” These muscle contractions release heat, which helps the body temperature to rise.

33 If body temperature rises too far above 37°C, the hypothalamus slows down cellular activities to reduce heat production. This is one of the reasons you may feel tired and sluggish on a hot day. The body also responds to high temperatures by producing sweat, which helps to cool the body surface by evaporation.

Quick Lab Guided Inquiry

34 How Do You Respond to an External Stimulus?

1. Have your partner put on safety goggles.
2. Crumple up a sheet of scrap paper into a ball.
3. Watch your partner's eyes carefully as you toss the paper ball toward his or her face.
4. Repeat step 3, three times.
5. Exchange roles and repeat steps 1, 3, and 4.

ANALYZE AND CONCLUDE

1. **Observe** Describe your partner's reaction to step 3.
2. **Compare and Contrast** Did you see any change in behavior as you repeated step 3? Explain.
3. **Infer** What is the function of the blink reflex?
4. **Develop Models** Draw a model to show how different systems of the body interacted.

30 KEY IDEA

Your body regulates **temperature** the same way a thermostat does, through negative feedback. **Figure 27-4** shows both loops: too hot and too cold.

31 SENSOR · CONTROL CENTER

The **hypothalamus** (part of the brain) is the body's thermostat. Its nerve cells **sense** body temperature and send signals that start or stop warming and cooling.

32 TOO COLD LOOP

Too cold → hypothalamus → cells speed up **metabolism** and muscles **shiver** → heat released → temperature rises back to normal.

33 TOO HOT LOOP

Too hot → hypothalamus → cells slow down and you **sweat** → evaporation cools the skin → temperature falls back to normal. Each response opposes the change, so both loops are **negative feedback**.

34 CONNECTION

Blinking when something flies at your face is a **response to an external stimulus**, another characteristic of life (p. 24).

HS-LS1-2

Two more negative feedback loops: water balance and blood sugar.

35 KEY IDEA

Hormone systems are also controlled by **negative feedback**: a rise in something “feeds back” to shut the system down.

36 WATER BALANCE LOOP

Sweating → water lost → blood too concentrated → hypothalamus → ADH (kidneys **save** water) and **thirst** (you drink) → water back to normal. Drinking a lot → less ADH → kidneys remove more water. Upper **and** lower limits = a **range**.

37 BLOOD SUGAR LOOP

After a meal: glucose **high** → pancreas releases **insulin** → cells store glucose → level falls. Between meals: glucose **low** → **glucagon** → liver releases glucose → level rises. Two opposing hormones keep it in range.

38 RANGE

“Within a normal **range**”: homeostasis doesn’t keep glucose at one exact number. It keeps it between limits (about 70–110 mg per 100 mL of blood).

39 READING CHECK: ANSWER

When blood glucose **rises**, the pancreas releases **insulin**, which makes cells take up and store glucose. When it **falls**, the pancreas releases **glucagon**, which makes the liver release glucose. Both bring the level back to normal.

Control of the Endocrine System The endocrine system is one of the master control systems of the body. Like most body systems, the endocrine system is regulated by negative feedback mechanisms that function to maintain homeostasis. Recall that negative feedback, also called feedback inhibition, occurs when an increase in a substance “feeds back” in a way that inhibits the system. Concentrations of hormones, and their effects on other body systems, are controlled in similar ways.

BUILD VOCABULARY

Prefixes The prefixes *anti-* and *ante-* can be easily confused. *Anti-*, as in *antidiuretic*, means “against.” *Ante-*, as in *anterior*, means “before.”

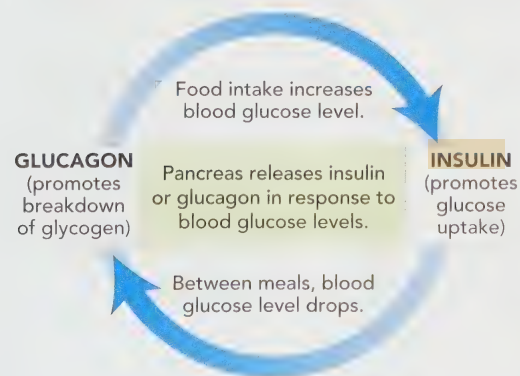
36 Maintaining Water Balance Water balance in the body is one example of how the endocrine system maintains homeostasis. When you exercise, you sweat and lose water. If this water loss continued, your body would soon become dehydrated. But your hypothalamus contains cells that monitor blood water content. As you lose water, concentrations of dissolved materials in blood rise. The hypothalamus responds in two ways. First, it signals the posterior pituitary gland to release antidiuretic hormone (ADH). ADH molecules are carried to the kidneys, where they slow the removal of water from blood. Second, the hypothalamus generates nerve impulses that produce a sensation of thirst to which we respond if water is available.

When concentrations of dissolved materials fall, as it might after you drink several glasses of water, the pituitary releases less ADH. As ADH concentrations fall, the kidneys remove more water from the blood, restoring its proper concentration. This homeostatic system sets both upper and lower limits for blood water content.

37 Blood Glucose Regulation Glucose concentration in the bloodstream is controlled by insulin and glucagon. When blood glucose concentration rises, the pancreas releases insulin. Insulin stimulates liver and skeletal muscle cells to convert blood glucose to glycogen, and it stimulates fat cells to convert glucose to lipids. These actions together prevent blood glucose concentrations from rising too rapidly. They also store energy for future use. In between meals, when blood glucose concentration drops, the pancreas releases glucagon. Glucagon stimulates liver and skeletal muscle cells to break down glycogen and release glucose into the blood. These actions raise blood glucose concentrations back to normal. Figure 27-27 shows this feedback loop.

Figure 27-27
Glucagon and Insulin

38 Insulin and glucagon are opposing hormones that ensure that blood glucose levels stay within a normal range.



READING CHECK

39 Summarize How does the endocrine system respond to rising or falling blood glucose levels?

READING TOOL

Make a chart that lists and describes the different steps of scientific methodology.



INTERACTIVITY

Try your hand at a simulation of a scientific investigation.

Observing and Asking Questions Scientific investigations begin with **observation**, the act of noticing and describing events or processes in a careful, orderly way. Of course, scientific observation involves more than just “looking.” The right kind of observation leads to asking questions that no one has asked (or answered) before.

Inferring and Forming a Hypothesis After posing questions, scientists use further observations to make inferences. An **inference** is a logical interpretation based on what scientists already know. Inference, combined with a creative imagination, can lead to a hypothesis. A **hypothesis** is a tentative scientific explanation that can be tested by further observation or by experimentation. Scientific hypotheses must then be tested by gathering data that can either support or reject them.

41 Designing Controlled Experiments Testing hypotheses often involves designing experiments that measure factors that can change. These changeable factors are called variables. Some possible variables include temperature, light, time, and availability of nutrients. Ideally, a hypothesis should be tested by an experiment in which only one variable is changed. All other variables should be kept unchanged, or controlled. This is called a **controlled experiment**.

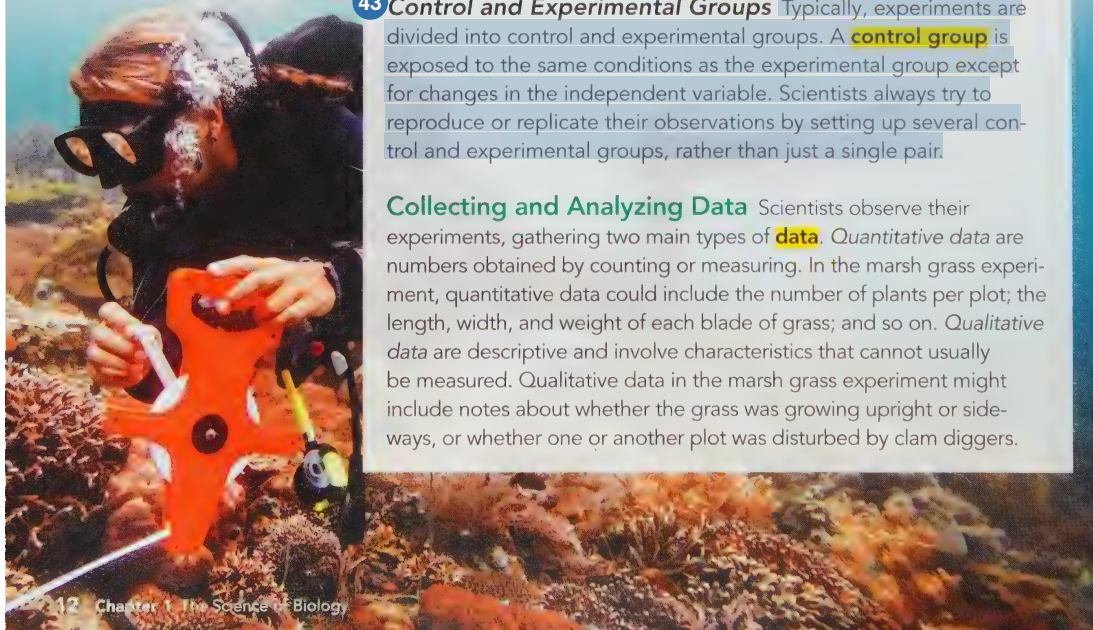
42 Controlling Variables It is important to control variables because if several variables are changed, researchers can’t easily tell which variable is responsible for any results they observe. The variable deliberately changed is called the **independent variable**. The variable that is observed and that changes in response to the independent variable is called the **dependent variable**.

43 Control and Experimental Groups Typically, experiments are divided into control and experimental groups. A **control group** is exposed to the same conditions as the experimental group except for changes in the independent variable. Scientists always try to reproduce or replicate their observations by setting up several control and experimental groups, rather than just a single pair.

Collecting and Analyzing Data Scientists observe their experiments, gathering two main types of **data**. **Quantitative data** are numbers obtained by counting or measuring. In the marsh grass experiment, quantitative data could include the number of plants per plot; the length, width, and weight of each blade of grass; and so on. **Qualitative data** are descriptive and involve characteristics that cannot usually be measured. Qualitative data in the marsh grass experiment might include notes about whether the grass was growing upright or sideways, or whether one or another plot was disturbed by clam diggers.

Figure 1-3
Collecting Data

This scientist is collecting data on a coral reef in Indonesia.



Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 12

Back to scientific investigation, the first half of the 1B target. This page defines the parts of a controlled experiment.

40 HYPOTHESIS

A **hypothesis** is a testable explanation. Data from experiments either **support** or **reject** it; they never “prove” it.

41 CONTROLLED EXPERIMENT

A **controlled experiment** changes **one** variable and keeps everything else the same, so any difference in results must come from that one variable.

42 INDEPENDENT · DEPENDENT VARIABLE

Independent variable: what *you* change on purpose (amount of nitrogen). **Dependent variable:** what you measure, which *depends* on it (grass height).

43 EXPERIMENTAL CONTROL

The **control group** gets the same conditions minus the independent variable. It’s the comparison that shows whether the variable had any effect. Using several groups (**replication**) makes results more reliable.

44 A 1B-STYLE EXPERIMENT

Does a sports drink help keep body temperature stable during exercise? **Independent variable:** drink (sports drink or water). **Dependent variable:** body temperature after 20 minutes of exercise. **Control group:** the water group. **Keep constant:** exercise, room temperature, time, and amount drunk.

45 Developing and Using Models Both scientists and engineers use models, such as those shown in **Figure 1-10**. When you hear the word *model*, you may think of model trains or other representations of much larger objects. But there are many different kinds of models. Mathematical representations and computer simulations are examples, as are two-dimensional drawings, diagrams, and maps. In fact, the flowcharts used throughout this lesson are models. Models help visualize and summarize ideas and communicate ideas to others.

Using Mathematics and Computational Thinking

Mathematics and computational thinking (the process of mathematical calculation) are also important tools. The relationships between variables are essential to the understanding of scientific phenomena. Ratios, rates, percentages, and unit conversions are used to analyze and interpret data. A mathematical representation can represent and model data; it can also support claims and explanations.

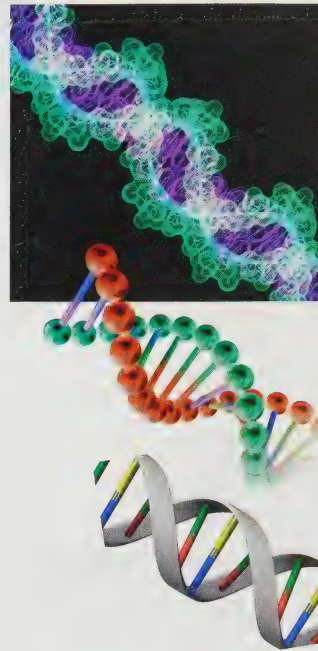
Constructing Explanations and Designing Solutions

Scientists aim to construct explanations for events in the natural world that caused them to ask a question. After a scientist collects data, analyzes and interprets that data, and finds that it supports the hypothesis, the next step is to construct an explanation. **46** An explanation describes how variables relate to one another, and it is supported by evidence. In engineering, the goal is to design a solution to a problem within certain constraints.

47 Engaging in Argument From Evidence In science, an argument is a reason or set of reasons used to persuade others that an idea is right or wrong. Once a scientist has constructed an explanation, the next step is to use evidence (analyzed and interpreted data) to persuade others that the explanation is correct. Part of this process is receiving critiques and responding thoughtfully. Engineers defend claims that a design solution is effective or evaluate competing design solutions.

Figure 1-10
Models

Models can be elaborate pictures that a computer generates (top), physical models (middle), or illustrations (bottom). All three of these models show DNA, a molecule essential for life.



Lesson 1.2 Science in Context

Miller & Levine Biology, textbook page 21

Models, explanations, and arguments: the methods named in the 1B target (“explanations ... revised through argumentation”).

45 MODEL

A **model** can be a drawing, flowchart, physical object, equation, or computer simulation. Your feedback-loop diagrams and the thermostat are both models of homeostasis.

46 KEY IDEA

An **explanation** tells *how variables relate* and is backed by evidence. Example: “sweating lowers body temperature because evaporation removes heat.”

47 ARGUMENT FROM EVIDENCE

This is what the 1B target means by **revised through argumentation**: scientists use evidence to persuade, listen to critiques, and improve their explanations. CER (claim, evidence, reasoning) is how you practice it.

48 REVIEW 1–2: ANSWERS

1. Curiosity leads scientists to ask new questions; skepticism makes them demand evidence before accepting an answer. 2. Peer reviewers find mistakes and weak spots, so scientists can fix their methods and explanations before (and after) publishing.

49 REVIEW 3–6: ANSWERS

3. Science affects society (health, environment, laws), and society’s needs shape which questions science studies. 4. Testing ideas is how scientists and engineers find out whether an explanation or design actually works. 5. Be skeptical: the claim hasn’t been checked by independent experts, so there’s no reliable evidence yet that the drink works. 6. The researcher works for the company that sells the pesticide, so the study could be **biased**. Independent scientists should repeat it.

LESSON 1.2 Review

KEY QUESTIONS

- 48** 1. How are both curiosity and skepticism useful in science?
2. How can peer review help scientists improve their work?
3. How is the use of science related to its context in society?
4. Why is testing ideas an important part of all science and engineering practices?

CRITICAL THINKING

- 49** 5. **Apply Concepts** An advertisement claims that studies of a new sports drink show it boosts energy. You discover that none of the study results have been peer-reviewed. What would you tell consumers who are considering buying this product?
6. **Apply Concepts** A study shows that a new pesticide is safe for use on food crops. The researcher who conducted the study works for the pesticide company. What potential biases may have affected the study?

Part 1 (continued). Other sources for this target

The class textbook pages are reproduced in full above. These excerpts cover the rest: the reference book *Campbell Biology* and anything the class textbook leaves out.

Campbell Biology

Properties of life (Ch. 1, p. 3). Campbell says life “defies a simple definition. We recognize life by what living things do.” Its Figure 1.2 lists seven properties: **order, evolutionary adaptation, regulation, reproduction, energy processing, growth and development, and response to the environment.**

Feedback regulation (Ch. 1, p. 9)

In feedback regulation, the output or product of a process regulates that very process. The most common form of regulation in living systems is negative feedback, a loop in which the response acts in opposition to the initial stimulus. ... Though less common than processes regulated by negative feedback, there are also many biological processes regulated by **positive feedback, in which an end product reinforces or strengthens the stimulus.** The clotting of your blood in response to injury is an example.

Metabolism (Ch. 8, p. 146)

The totality of an organism’s chemical reactions is called metabolism (from the Greek *metabole*, change). ... Some metabolic pathways release energy by breaking down complex molecules to simpler compounds. These degradative processes are called catabolic pathways, or breakdown pathways. ... Anabolic pathways, in contrast, consume energy to build complicated molecules from simpler ones; they are sometimes called biosynthetic pathways.

Homeostasis and the normal range (Ch. 40, p. 893)

In achieving homeostasis, animals maintain a “steady state”—a relatively constant internal environment—even when the external environment changes significantly. ... For example, humans maintain a fairly constant body temperature of between approximately 36° and 38°C (97°–99°F), a blood pH within 0.1 pH unit of 7.4, and a blood glucose concentration that is predominantly in the range of 70–110 mg of glucose per 100 mL of blood.

Dynamic equilibrium and positive feedback (Ch. 40, p. 894)

Homeostasis is a dynamic equilibrium, an interplay between external factors that tend to change the internal environment and internal control mechanisms that oppose such changes. ... Unlike negative feedback, positive feedback is a control mechanism that amplifies the stimulus. In animals, positive-feedback loops do not play a major role in homeostasis, but instead help drive processes to completion. During childbirth, for instance, the pressure of the baby’s head against sensors near the opening of the mother’s uterus stimulates the uterus to contract.

Part 2. Summary

The big idea of 1B. Scientists answer “What is life?” and “How do living things stay alive?” with evidence from many methods: observation, controlled experiments, and models. They keep revising their explanations when new evidence or better arguments appear. Two ideas sit at the center:

1. What counts as alive. No single trait defines life. Fire moves and gives off light, but it isn’t alive. Instead, biologists use a *set of criteria*. According to Miller & Levine, living things:

1. are made of **cells**
2. **reproduce** (sexually or asexually)

3. are based on a universal **genetic code** (DNA)
4. **grow and develop**
5. **obtain and use materials and energy** (metabolism)
6. **respond to their environment** (stimuli)
7. **maintain a stable internal environment** (homeostasis)
8. **evolve**, changing over time as a group

Something is considered alive when it shows *all* of these, not just one or two. **Viruses** are the classic borderline case: they reproduce and evolve, but only inside a host cell, and they aren't made of cells. Whether to call them alive is a real scientific argument, which is exactly what the 1B target statement describes.

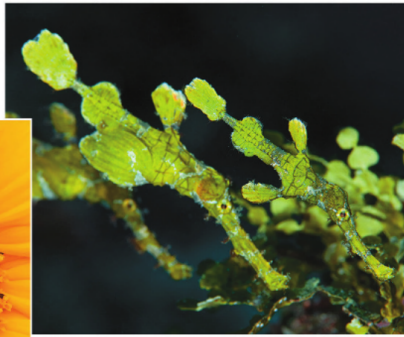
2. How living things stay stable.

- **Metabolism** is all the chemical reactions in an organism. Some build molecules up (photosynthesis, protein synthesis), and some break them down (cellular respiration, digestion).
- **Homeostasis** is keeping internal conditions (temperature, water, blood sugar, pH) within a safe **range** even when the outside world changes. It is a **dynamic equilibrium**: conditions constantly drift and get corrected, rather than staying perfectly fixed.
- Organisms maintain homeostasis through **feedback loops**, where a change in a condition triggers a response that affects that same condition.
 - **Negative feedback** *opposes* the change, which keeps the system stable. Examples: sweating and shivering for body temperature, insulin and glucagon for blood sugar. This is how homeostasis works.
 - **Positive feedback** *amplifies* the change, pushing the system further in the same direction until something ends it. Examples: childbirth contractions, blood clotting, the melting of ice speeding up global warming. It does *not* maintain homeostasis, and in nature it's often harmful.

One-sentence version: life is recognized by a set of shared characteristics, and one of the most important is homeostasis, which organisms maintain mostly through negative feedback loops that keep conditions within a stable range.

Figure 1.2 Some properties of life.

▼ **Order.** This close-up of a sunflower illustrates the highly ordered structure that characterizes life.



▲ **Evolutionary adaptation.** The overall appearance of these ghost pipefish camouflages the animals in their environment. Such adaptations evolve over countless generations by the reproductive success of those individuals with heritable traits that are best suited to their environments.



▲ **Regulation.** The regulation of blood flow through the blood vessels of this jackrabbit's ears helps maintain a constant body temperature by adjusting heat exchange with the surrounding air.



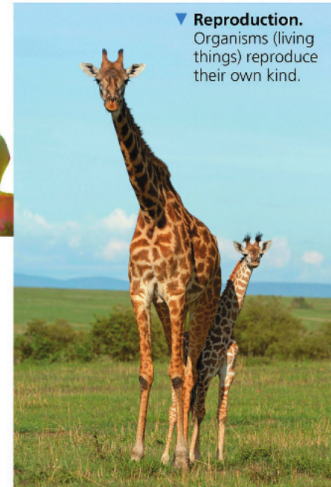
▲ **Energy processing.** This butterfly obtains fuel in the form of nectar from flowers. The butterfly will use chemical energy stored in its food to power flight and other work.



▲ **Growth and development.** Inherited information carried by genes controls the pattern of growth and development of organisms, such as this oak seedling.



▲ **Response to the environment.** The Venus flytrap on the left closed its trap rapidly in response to the environmental stimulus of a grasshopper landing on the open trap.



▼ **Reproduction.** Organisms (living things) reproduce their own kind.

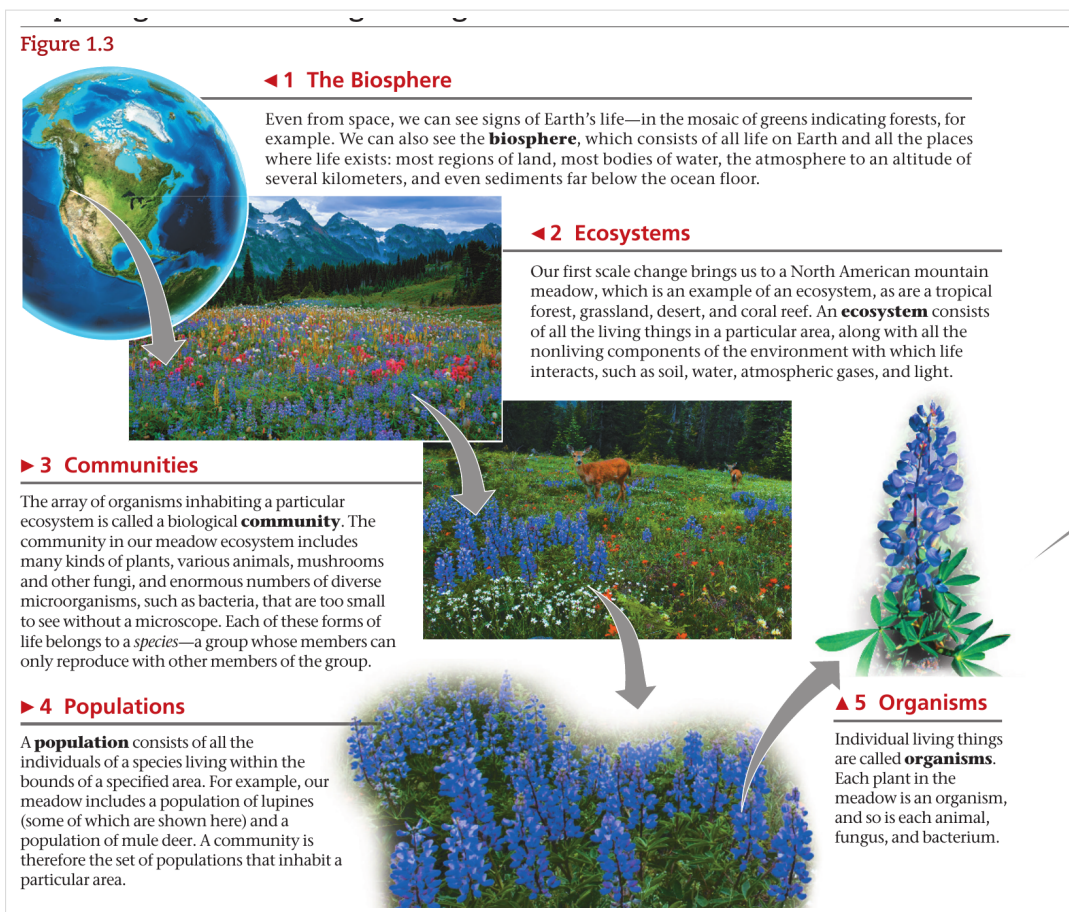
Some properties of life: order, evolutionary adaptation, regulation (homeostasis), reproduction, energy processing, growth and development, and response to the environment.

Source: Campbell Biology, Figure 1.2, p. 3

Part 3. Biology vocabulary

organism

- **Definition:** an individual living thing. "Living things, or organisms, are made up of small, self-contained units called cells" (M&L p. 23).
- **Unicellular vs. multicellular** (from the What Is Life? notes): a unicellular organism is a single cell (bacteria, amoeba). A multicellular organism is made of many cells (a tree, a human).
- **Examples:** a bacterium, a mushroom, a sunflower, a person. **Non-examples:** a virus (borderline, usually not counted), a cell taken out of your body (alive, but not an organism by itself), a fire.
- **Link to ecology:** organisms are the smallest level in ecology. Organisms → population → community → ecosystem → biosphere.

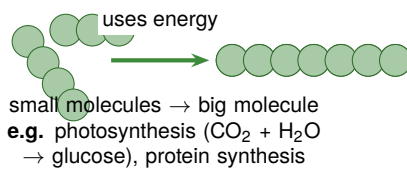
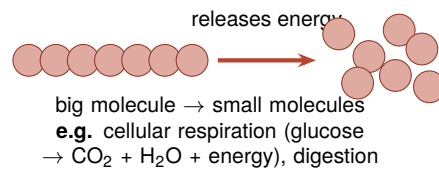


Levels of biological organization, zooming in from the biosphere to one organism. An organism is an individual living thing; many organisms of one species form a population.

Source: Campbell Biology, Figure 1.3, p. 4

metabolism

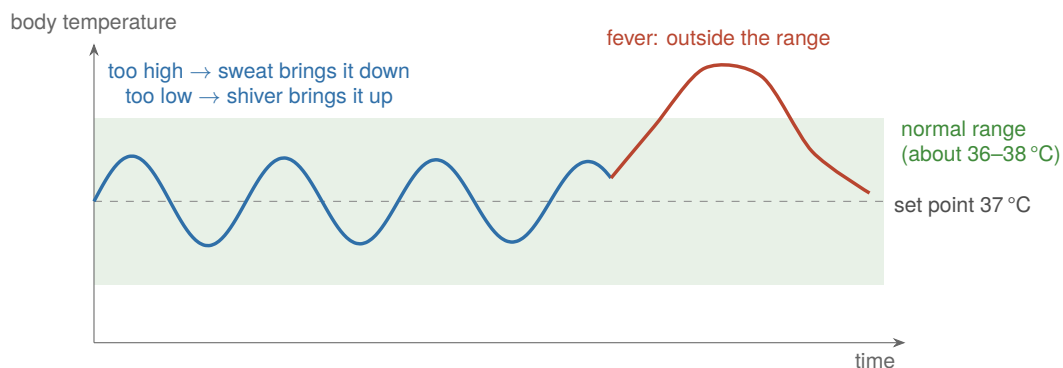
- **Definition:** “The combination of chemical reactions [an organism uses] as it carries out its life processes” (M&L p. 24). Campbell (p. 146): “The totality of an organism’s chemical reactions.” The teacher’s definition: “All of the chemical reactions that build up and break down molecules in living things.”
- **Two directions:**
 - | Building up (anabolic) | Breaking down (catabolic) | |—|—| | uses energy | releases energy | | small molecules → big molecules | big molecules → small molecules | | photosynthesis ($\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose}$) | cellular respiration ($\text{glucose} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{energy}$) | | protein synthesis (amino acids → protein) | digestion (starch → sugars) |
- **Memory hook:** *meta* + *bol* comes from the Greek *metabole*, “change” (Campbell p. 146). Metabolism is all the chemical *changes* in your body.
- **Why it matters for 1C:** photosynthesis and cellular respiration are how energy and matter move through ecosystems.

Building up (anabolic)**Breaking down (catabolic)**

Metabolism runs in two directions. Energy released by breaking molecules down powers the reactions that build molecules up.

homeostasis

- **Definition:** “the relatively constant internal conditions that organisms maintain despite changes in internal and external environments” (M&L p. 907). The teacher’s note: “Organisms use feedback mechanisms to maintain internal conditions within certain limits, even as external conditions change.”
- **Word parts:** *homeo-* = same, *-stasis* = “a state of balance or equilibrium” (M&L p. 907). Together: “keeping things the same.”
- **What gets kept stable in humans** (Campbell p. 893):
 - body temperature: about 36–38 °C (97–99 °F)
 - blood pH: within 0.1 of 7.4
 - blood glucose: mostly 70–110 mg per 100 mL
 - water balance
- **Key point:** homeostasis doesn’t mean *no* change. The value moves up and down within a range, which is why it’s called a **dynamic equilibrium** (Campbell p. 894; teacher’s slide 2).
- **Why it matters:** “any breakdown of homeostasis may have serious or even fatal consequences” (M&L p. 25). Heatstroke, hypothermia, dehydration, and uncontrolled diabetes are all failures of homeostasis.



Homeostasis is a dynamic equilibrium. Body temperature wobbles but stays within the normal range around the set point. Only when feedback fails (here, a fever) does it leave the range.

stable / stability

- **Definition:** *stable* means staying about the same and resisting change. *Stability* is the quality of being stable.

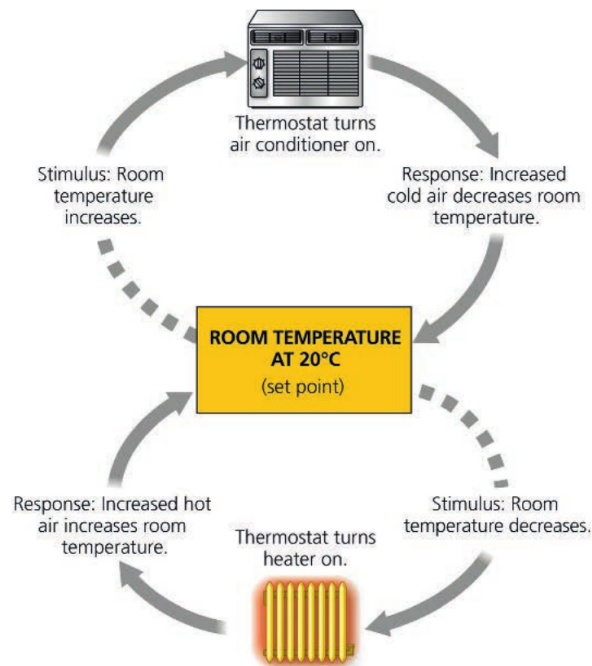
- **In biology:** a stable internal environment is one of the characteristics of life. “Systems controlled by feedback inhibition are generally very stable” (M&L p. 907).
- **Stable $\neq$ unchanging:** a stable body temperature still rises a little during exercise and falls a little at night. It just returns toward the normal value.
- **Crosscutting concept:** “Stability and Change” is one of the big themes of biology (M&L p. 25). Later in this unit you’ll see it again: stable ecosystems (2A) and stable populations (2C).

feedback loop

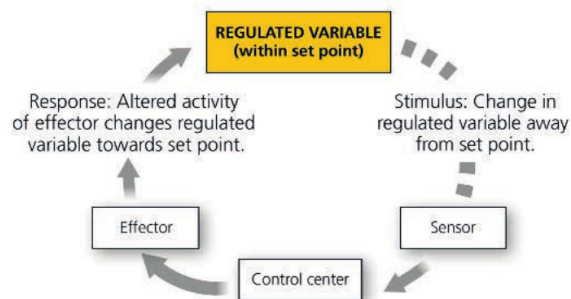
- **Definition:** “any mechanism in which a condition triggers some action that causes a change in that same condition” (teacher’s slide 2). Campbell (p. 9): “the output or product of a process regulates that very process.”
- **Why it’s called a loop:** the result circles back to affect the starting point.

condition changes → **sensor detects it** → **control center** → **response** → **the condition changes back**
(and the loop starts again)

- **Parts** (Campbell pp. 893–894; M&L p. 907):
 - **Stimulus / condition:** the thing being regulated, such as body temperature
 - **Sensor:** detects the change, such as nerve cells in the hypothalamus or a thermostat
 - **Set point:** the target value, such as 37 °C
 - **Effector / response:** what acts to change the condition, such as sweat glands, muscles (shivering), or a furnace
- **Two types:** negative and positive (below).

Figure 40.8 Negative feedback loops and homeostasis.

(a) A nonliving example of temperature regulation. Thermostat components monitor room temperature and compare it with the set point. If the room temperature changes away from the set point, the thermostat activates the heater or the air conditioner to reverse the change.



(b) Components of a generalized negative feedback loop.

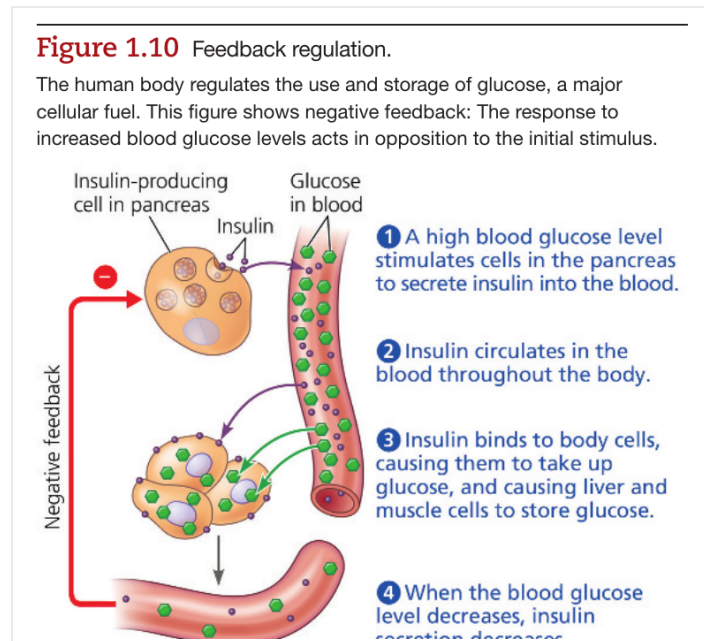
(a) A thermostat keeps room temperature near its set point. (b) Every negative feedback loop has the same parts: a sensor, a control center, and an effector whose response pushes the variable back toward the set point.

Source: Campbell Biology, Figure 40.8, p. 894

negative feedback

- **Definition:** “the process in which a stimulus produces a response that opposes the original stimulus” (M&L p. 907). Also called **feedback inhibition**.
- **What it does:** it *reverses* a change and brings the condition back toward the set point. It keeps things **stable**, so it’s the main way organisms maintain homeostasis. It is “the most common form of regulation in living systems” (Campbell p. 9).
- **Examples:**
 - **Body temperature** (teacher’s slide 4; M&L p. 908): too hot → sweat → cools down. Too cold → shiver → warms up.

- **Blood glucose** (M&L p. 932; Campbell p. 9): high after a meal → pancreas releases insulin → cells take up and store glucose → level falls. Low between meals → glucagon → liver releases glucose → level rises.
- **Water balance** (M&L pp. 24, 932): water is low → you feel thirsty and the kidneys save water (ADH) → water returns to normal.
- **Thermostat** (M&L p. 907): too cold → furnace on → warms → furnace off.
- **Climate** (teacher's slide 9): Earth warms → more evaporation → more clouds → more sunlight reflected → cooling.
- **Memory hook:** negative = *negates* (cancels) the change.



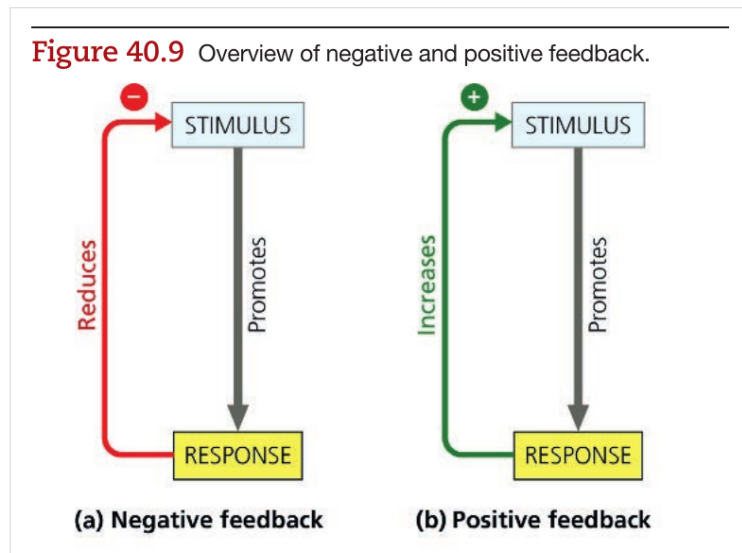
Negative feedback for blood sugar: high glucose → the pancreas releases insulin → cells take up glucose → glucose falls → insulin release slows.

Source: Campbell Biology, Figure 1.10, p. 10

positive feedback

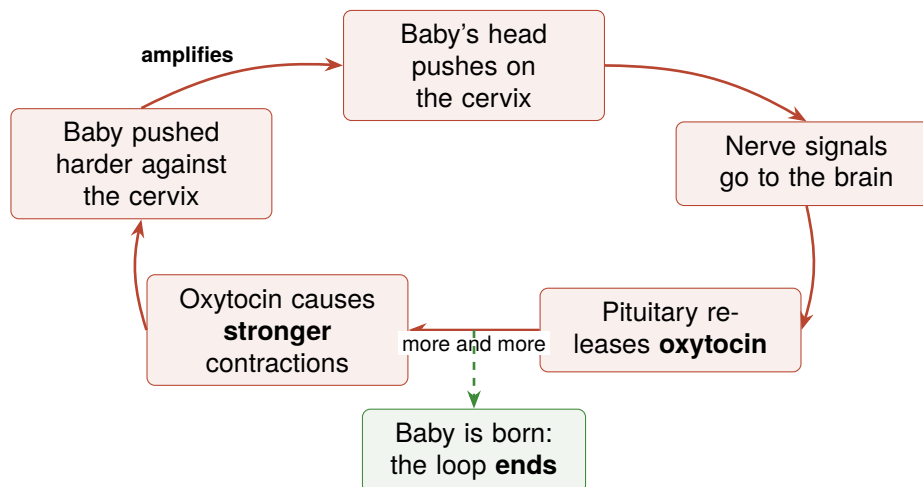
- **Definition:** “an end product reinforces or strengthens the stimulus” (Campbell p. 9). “A control mechanism that amplifies the stimulus” (Campbell p. 894).
- **What it does:** it pushes the condition *further in the same direction*, so the change keeps growing until something outside the loop stops it. It does **not** maintain homeostasis. In animals it helps “drive processes to completion” (Campbell p. 894). In natural systems it is often harmful (teacher's slide 3).
- **Examples:**
 - **Childbirth** (teacher's slide 5; Campbell p. 894): baby's head presses on the cervix → nerve signals to the brain → pituitary releases oxytocin → stronger contractions → more pressure → even more oxytocin ... until the baby is born.
 - **Blood clotting** (Campbell p. 9): platelets gather at a wound and release chemicals that attract more platelets, until the clot seals the wound.
 - **Ice-albedo** (teacher's slide 7): warming melts ice → darker surface absorbs more energy → more warming → more melting.

- **Permafrost** (slide 8) and **soil decomposition** (slide 10): warming releases more greenhouse gases → more warming.
- **Memory hook:** positive = *adds to* the change. “Positive” doesn’t mean “good.” It means “in the same direction.”



(a) Negative feedback: the response reduces the stimulus. (b) Positive feedback: the response increases the stimulus.

Source: Campbell Biology, Figure 40.9, p. 894



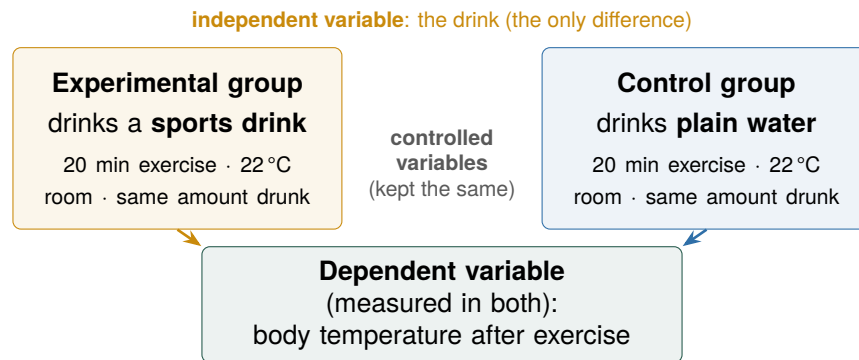
Positive feedback in childbirth (teacher's slide 5): each step makes the next one stronger, until the baby is born and the loop stops.

experimental control

- **Definition:** the part of an experiment that gives you something to compare against. It usually means the **control group**, which “is exposed to the same conditions as the experimental group except for changes in the independent variable” (M&L p. 12). It also refers to keeping all other variables **controlled** (held constant).
- **Why it matters:** without a control you can’t tell whether your independent variable caused the result or something else did. In the salt marsh experiment, the no-nitrogen plots were the control (M&L p.

11).

- **Example tied to 1B:** to test whether a sports drink helps keep body temperature stable during exercise, one group drinks the sports drink (experimental). The control group drinks plain water, with the same exercise, room temperature, and time. Only the drink differs.
- **Don't confuse:** a *control group* (the comparison group) with *controlled variables* (factors kept the same in both groups). Both are part of experimental control.



A controlled experiment. The two groups differ in only one thing, the independent variable, so any difference in the result must come from it.

Part 4. Academic vocabulary

condition

- **Meaning:** the state something is in, or a factor in the surroundings. *Internal conditions* are inside the body (temperature, water content). *External conditions* are outside (air temperature, sunlight).
- **In 1B:** “most organisms must keep internal conditions, such as temperature and water content, fairly constant to survive” (M&L p. 24). In a feedback diagram, the **condition** is the box in the middle that’s being regulated (teacher’s slide 4).
- **In experiments:** “a control group is exposed to the same *conditions* as the experimental group” (M&L p. 12).

constant

- **Meaning (adjective):** staying the same, not changing. (**noun**): something that stays the same, such as “the constants in an experiment.”
- **In 1B:** homeostasis keeps conditions “fairly constant” (M&L p. 24) or “relatively constant” (M&L p. 907). The words *fairly* and *relatively* matter, because conditions move within a range rather than being perfectly fixed.
- **In experiments:** controlled variables are held *constant*.

criteria

- **Meaning:** standards used to judge or decide something. *Criteria* is plural; the singular is *criterion*.
- **In 1B:** “What are the **criteria** to determine if something is alive?” The criteria are the eight characteristics of life. A thing must meet *all* of them to be classified as living.

- **Example sentence:** “Fire meets some criteria for life, such as using energy and growing, but it does not meet the criteria of being made of cells or having a genetic code.”

dynamic

- **Meaning:** always changing or active. The opposite of *static*.
- **In 1B:** homeostasis is a **dynamic equilibrium**, meaning balance kept by constant adjustment (teacher’s slide 2; Campbell p. 894). The textbook also calls the process of science “dynamic” because its steps loop back on themselves (M&L p. 15).
- **Picture it:** a driver keeps “slightly moving the wheel left and right” to stay in the lane (M&L p. 907). The car stays centered, but only because of constant small corrections. That’s dynamic stability.

equilibrium

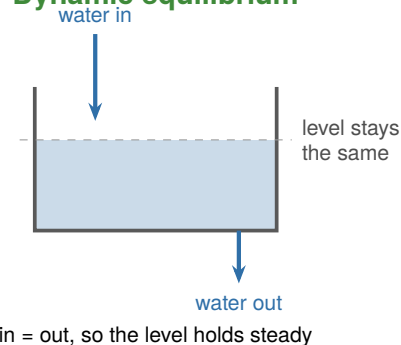
- **Meaning:** a state of balance, where opposing forces or processes cancel each other out.
- **In 1B:** the root *stasis* in homeostasis means “a state of balance or equilibrium” (M&L p. 907). In **dynamic equilibrium**, things keep moving (you eat and excrete every day), but the overall level stays balanced (your weight stays within a few pounds; teacher’s slide 2).
- **Static vs. dynamic equilibrium:** a book resting on a table is static equilibrium (nothing moving). Your body temperature is dynamic equilibrium (constant heating and cooling that balance out).

Static equilibrium

a book resting on a table:
nothing moves



Dynamic equilibrium



Static equilibrium: nothing moves. Dynamic equilibrium: things keep moving, but inputs and outputs balance, so the level stays the same, like body temperature or body weight.

mechanism

- **Meaning:** the process or set of parts by which something happens; *how* it works.
- **In 1B:** “The way organisms maintain homeostasis is through **feedback mechanisms**” (teacher’s slide 2). Shivering and sweating are the mechanisms for body temperature. M&L’s crosscutting concept “Cause and Effect: **Mechanism** and Explanation” (p. 25) means that science explains events by finding the mechanism that causes them.
- **Example sentence:** “Sweating is a cooling mechanism: evaporation of sweat removes heat from the skin.”

model

- **Meaning:** a simplified representation of an object, system, or idea. It can be physical, a drawing, a diagram, a flowchart, a math equation, or a computer simulation (M&L p. 21).
- **In 1B:**
 - The **feedback loop diagram** you draw in the Feedback practice is a model.
 - The **thermostat** is used as a model for homeostasis, “a nonliving system that keeps conditions within a certain range” (M&L p. 907).
- **Why scientists use models:** “Models help visualize and summarize ideas and communicate ideas to others” (M&L p. 21). All models leave something out. The thermostat model, for example, has one sensor, but the body uses many.

Figure 2.10 Covalent bonding in four molecules.

The number of electrons required to complete an atom's valence shell generally determines how many covalent bonds that atom will form. This figure shows several ways of indicating covalent bonds.

Name and Molecular Formula	Electron Distribution Diagram	Lewis Dot Structure and Structural Formula	Space-Filling Model
(a) Hydrogen (H₂). Two hydrogen atoms share one pair of electrons, forming a single bond.		H : H H — H	
(b) Oxygen (O₂). Two oxygen atoms share two pairs of electrons, forming a double bond.		:Ö :: Ö: O = O	
(c) Water (H₂O). Two hydrogen atoms and one oxygen atom are joined by single bonds, forming a molecule of water.		:Ö :: H H — O — H	
(d) Methane (CH₄). Four hydrogen atoms can satisfy the valence of one carbon atom, forming methane.		H H : C : H H — C — H H	

Four models of the same molecules: electron diagrams, structural formulas, and space-filling models. Each model shows some features and leaves others out. The thermostat in Figure 40.8 is a model too.

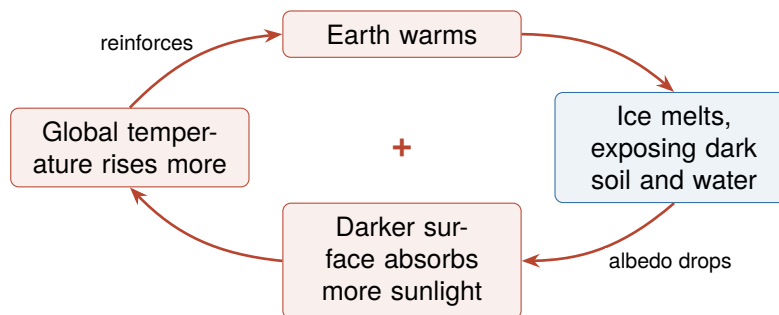
Source: Campbell Biology, Figure 2.10, p. 36

range

- **Meaning:** the span between a lower limit and an upper limit.
- **In 1B:** homeostasis keeps conditions “within a certain range, never allowing them to go too far one way or the other” (M&L p. 907). Normal human body temperature is about 36–38 °C; normal blood glucose is 70–110 mg/100 mL (Campbell p. 893).
- **Why it matters:** the teacher’s slide 2 says the stable state “is usually not entirely rigid, but can fluctuate within a certain range.” Going *outside* the range is when problems start, such as fever, hypothermia, or low blood sugar.

reinforce

- **Meaning:** to strengthen or add support to something.
- **In 1B:** positive feedback is when “an end product **reinforces** or strengthens the stimulus” (Campbell p. 9). The Feedback practice asks for “a **self-reinforcing** loop,” a loop that keeps strengthening itself.
- **Example sentence:** “Melting ice exposes dark water, which absorbs more heat and reinforces the warming.”
- **Everyday use:** practice reinforces learning (a 1A idea).



A self-reinforcing loop (teacher’s slide 7). Each step reinforces the warming that started it, so the change keeps growing: positive feedback.

Part 5. Study questions with answers

Try each one yourself first, then check.

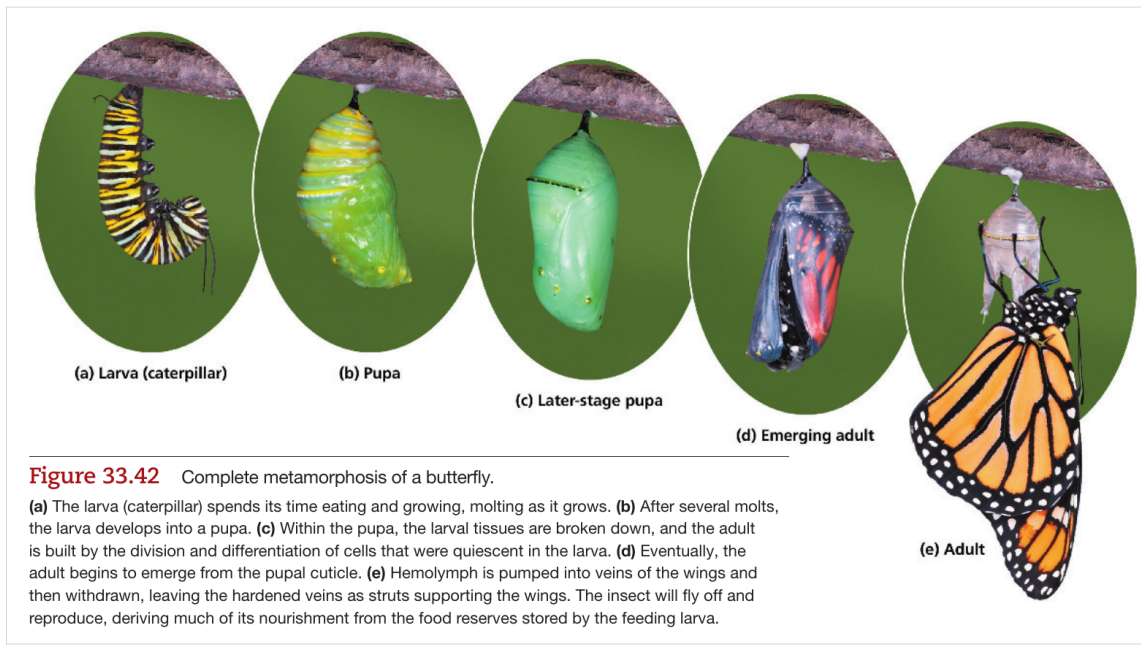
Q1. What are the criteria to determine if something is alive?

Short answer: No single trait is enough. A living thing shows *all* of these characteristics: it is made of cells, reproduces, has a universal genetic code (DNA), grows and develops, obtains and uses materials and energy, responds to its environment, maintains a stable internal environment (homeostasis), and evolves over time as a group (M&L p. 22).

Each criterion explained

#	Criterion	What it means	Example
1	Made of cells	Cells are “the smallest units of an organism that can be considered alive” (M&L p. 23). Organisms are unicellular (one cell) or multicellular (many cells).	Bacteria (1 cell); humans (trillions)
2	Reproduce	Organisms make new organisms. Sexual reproduction: two parents. Asexual reproduction: one parent (M&L p. 23).	Budding yeast (asexual); dogs (sexual)
3	Universal genetic code	Inherited instructions are carried in DNA, and the same code works in nearly every organism on Earth (M&L p. 23).	Offspring resemble their parents
4	Grow and develop	Organisms get bigger, and multicellular organisms <i>develop</i> : cells divide and differentiate into different types (M&L p. 23).	Egg → caterpillar → monarch butterfly
5	Obtain and use materials and energy	They take in matter and energy and use them in chemical reactions: metabolism (M&L p. 24).	Plants use sunlight; animals eat food
6	Respond to the environment	They detect and react to stimuli , both external (light, temperature) and internal (thirst) (M&L p. 24).	Sunflowers turning to follow the sun
7	Maintain a stable internal environment	Homeostasis: internal conditions stay within a safe range (M&L p. 24).	Sweating when hot
8	Evolve	A <i>group</i> of organisms changes over many generations. Individuals don't evolve (M&L p. 24).	Bacteria evolving antibiotic resistance

Campbell's version (p. 3) lists seven properties that overlap closely: order (organized structure), reproduction, growth and development, energy processing, response to the environment, regulation (homeostasis), and evolutionary adaptation.



Growth and development: a butterfly's complete metamorphosis, from larva (caterpillar) to pupa to adult. Development means changing form, not just getting bigger.

Source: Campbell Biology, Figure 33.42, p. 725

Why you need all of them (testing nonliving things):

- **Fire** grows, uses energy (burns fuel), gives off light, and “moves in its own way” (M&L p. 22). But it has no cells, no DNA, and no homeostasis, so it is **not alive**.
- **A car** moves, uses energy, and responds (to the driver), but has no cells, doesn't reproduce, and doesn't evolve. **Not alive**.
- **A crystal** grows, but has no cells or metabolism. **Not alive**.
- **A mule** is alive even though it's sterile (can't reproduce). It meets every other criterion and is made of cells from parents that did reproduce. The criteria describe living things *in general*, not every individual. (The What Is Life? notes ask: “Are there any living organisms that cannot reproduce?”)

Criterion	Oak tree	Bacterium	Fire	Car	Virus
1 Made of cells	✓	✓	×	×	×
2 Reproduces	✓	✓	×	×	?
3 Genetic code (DNA)	✓	✓	×	×	✓
4 Grows and develops	✓	✓	×	×	×
5 Uses materials and energy	✓	✓	✓	✓	×
6 Responds to environment	✓	✓	×	✓	×
7 Homeostasis	✓	✓	×	×	×
8 Evolves	✓	✓	×	×	?
Alive?	Yes	Yes	No	No	Border

Test each thing against all eight criteria. Only living things meet every one. ? = only inside a host cell.

Viruses: the borderline case (M&L p. 24; What Is Life? notes)

Viruses share with living things	Viruses lack
have genetic material (DNA or RNA)	not made of cells, just proteins, nucleic acids, and sometimes lipids
can reproduce, but <i>only</i> inside a host cell	have no metabolism of their own
evolve (e.g., new flu strains each year)	can't grow or maintain homeostasis
can regulate gene expression	can't reproduce on their own

Most biologists classify viruses as **nonliving** (or “at the border”), because they can’t do anything outside a host cell. This question has no single agreed answer, and scientists weigh the evidence and argue both sides. That is exactly the kind of “explanation revised through argumentation” the 1B target describes. For a test, state your position and support it with specific criteria, CER style.

Q2. Describe and give examples of metabolism.

Definition: Metabolism is **all of the chemical reactions that build up and break down molecules in living things** (teacher’s note). The class textbook calls it “the combination of chemical reactions [an organism uses] as it carries out its life processes” (M&L p. 24). Campbell calls it “the totality of an organism’s chemical reactions” (p. 146).

Why organisms need it. Just as “a building grows taller because workers use energy to assemble new materials” (M&L p. 24), organisms need *energy* and *materials* to grow, repair themselves, reproduce, and just stay alive. Metabolism is how they get energy out of food and use it to build what they need.

Two kinds of metabolic reactions (Campbell p. 146):

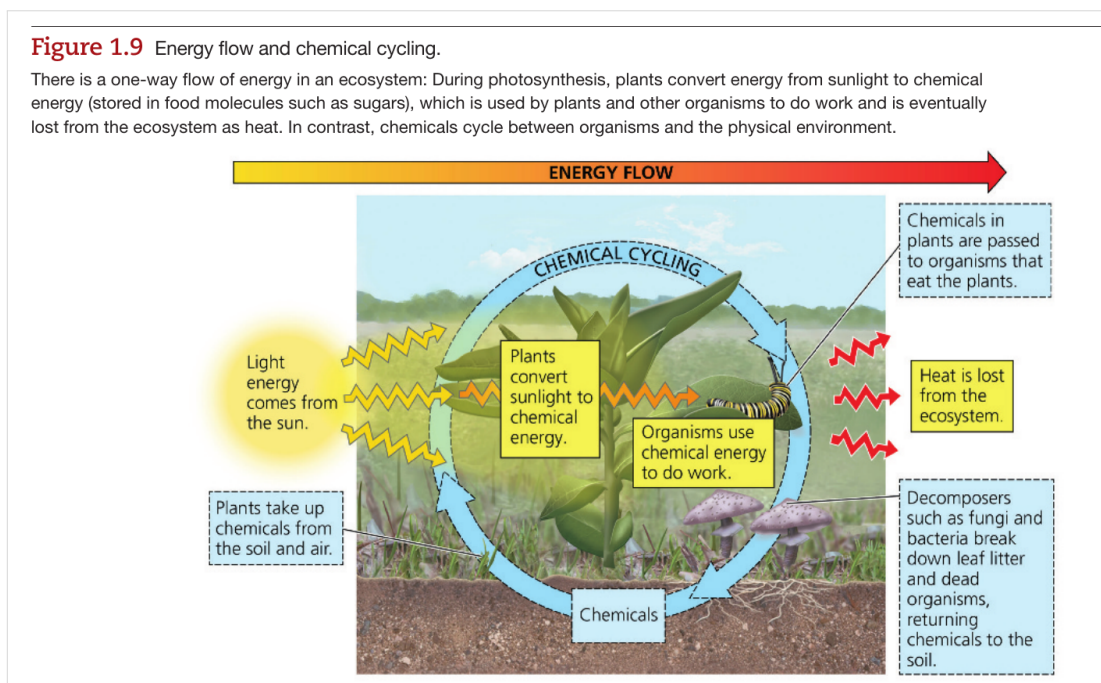
1. **Breaking down (catabolic):** big, complex molecules are broken into smaller ones, which **releases energy**.
2. **Building up (anabolic):** small molecules are assembled into bigger ones, which **uses energy**.

Energy released by breakdown reactions powers the building reactions.

Examples

Example	Build up or break down?	What happens	Where
Photosynthesis	Build up	Carbon dioxide + water + light energy → glucose + oxygen. Light energy is stored as chemical energy in sugar.	Plants, algae (chloroplasts)
Cellular respiration	Break down	Glucose + oxygen → carbon dioxide + water + energy (ATP) . This is the main way cells release usable energy.	Nearly all organisms (mitochondria)
Protein synthesis	Build up	Amino acids are joined into proteins, following instructions in DNA.	All cells (ribosomes)

Example	Build up or break down?	What happens	Where
Digestion	Break down	Large food molecules (starch, proteins, fats) → small molecules (sugars, amino acids, fatty acids) the body can absorb.	Digestive system
Storing glucose as glycogen	Build up	After a meal, insulin tells liver and muscle cells to link glucose into glycogen (M&L p. 932).	Liver, muscles
Breaking down glycogen	Break down	Between meals, glucagon tells the liver to release glucose (M&L p. 932).	Liver



Metabolism across a whole ecosystem: photosynthesis builds sugar using light energy, organisms break sugar down to do work, energy leaves as heat, and chemicals cycle back.

Source: Campbell Biology, Figure 1.9, p. 9

Connections:

- **To homeostasis:** metabolism produces heat. When you're cold, the hypothalamus signals cells to "speed up their activities," and the heat from "cellular respiration" raises body temperature (M&L p. 908). When you're hot, the body slows metabolism down.
- **To ecology (1C):** photosynthesis and cellular respiration are opposites, and together they move energy and carbon through ecosystems. Producers build sugar by photosynthesis; consumers break it down by respiration.

Q3. How do feedback mechanisms help to maintain homeostasis?

Short answer: Feedback mechanisms constantly detect changes in internal conditions and trigger responses that push the condition back toward normal. Mostly this happens through **negative feedback**,

which opposes a change. Because every drift away from the set point is corrected, conditions like temperature, water, and blood sugar stay within a safe **range**, which is homeostasis.

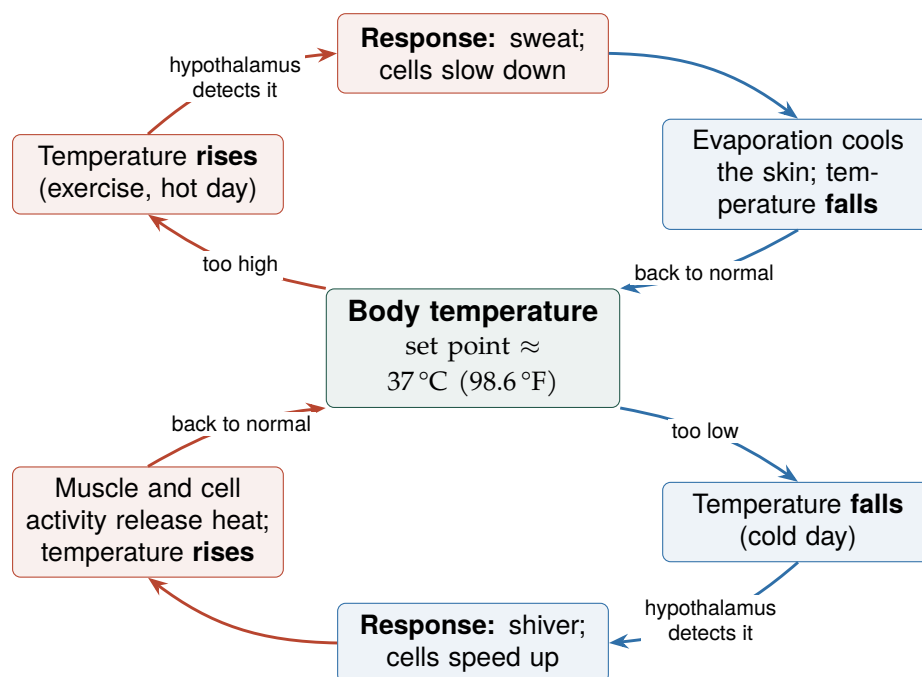
1. What homeostasis needs. Homeostasis means keeping internal conditions “relatively constant ... despite changes in internal and external environments” (M&L p. 907). The outside world keeps changing (hot days, meals, exercise), so the body needs a *control system* that keeps checking and correcting. That system is a **feedback loop**: “a condition triggers some action that causes a change in that same condition” (teacher’s slide 2).

2. How a negative feedback loop works. Every loop has the same parts (Campbell pp. 893–894):

- a **sensor** detects the condition
- a **control center** compares it to the **set point**
- an **effector** produces a **response** that pushes the condition the opposite way

Typically there are two loops working together: “mechanisms that lower a condition when it is too high, and other mechanisms that raise the level when it is too low” (teacher’s slide 2).

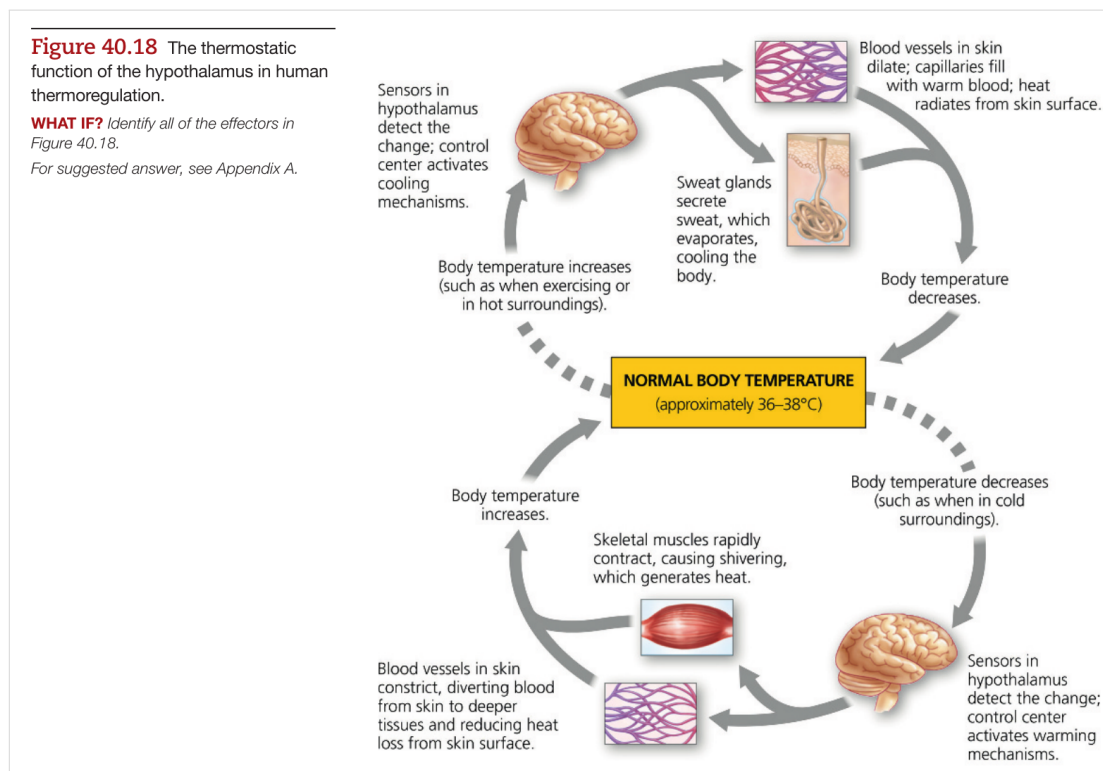
3. Example: body temperature (teacher’s slide 4; M&L p. 908)



Each response opposes the change that caused it, so both loops are negative feedback (teacher’s slide 4; M&L p. 908).

In words:

- **Too hot** (exercise, hot day): nerve cells in the hypothalamus detect it → sweat glands produce sweat, and cellular activity slows → evaporation cools the skin → temperature **falls** back toward 37°C.
- **Too cold**: the hypothalamus detects it → cells speed up metabolism, and muscles shiver → heat is released → temperature **rises** back toward 37°C.
- Each response **opposes** the change that triggered it, which is why it’s *negative* feedback. When the temperature is back in range, the response switches off, just like a thermostat turning off a furnace (M&L p. 907).

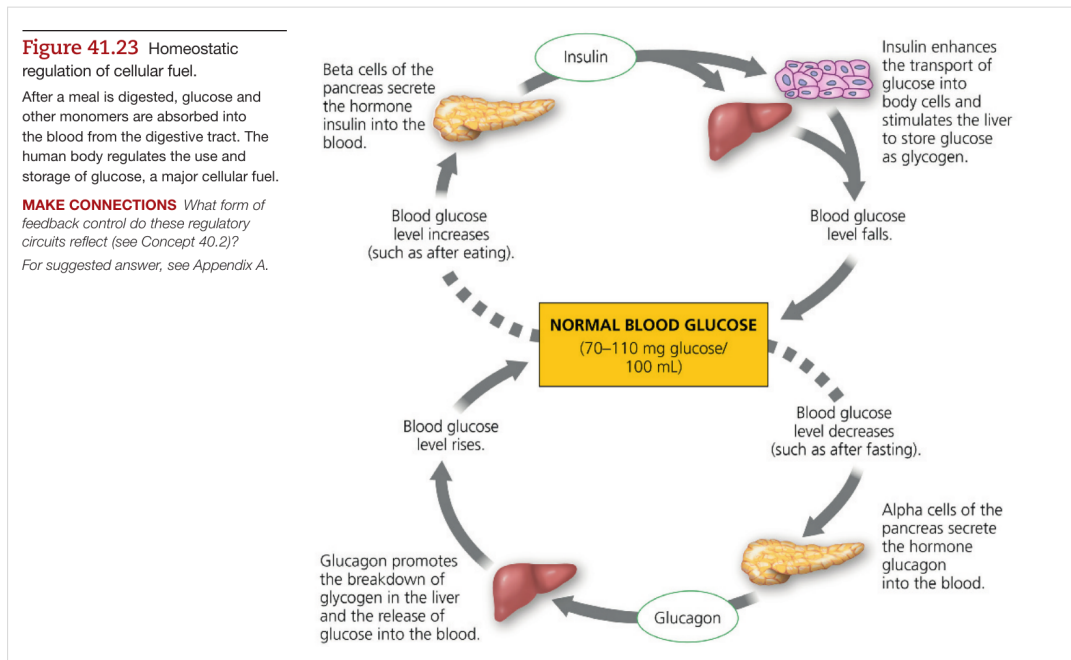


The hypothalamus works like a thermostat. Too warm: sweating and widened skin blood vessels cool the body. Too cold: shivering and narrowed blood vessels warm it.

Source: Campbell Biology, Figure 40.18, p. 901

4. Example: blood glucose (M&L p. 932; Campbell p. 9)

- **After a meal**, blood glucose rises → the pancreas releases **insulin** → liver and muscle cells take up glucose and store it as glycogen → blood glucose **falls**.
- **Between meals**, blood glucose falls → the pancreas releases **glucagon** → the liver breaks down glycogen and releases glucose → blood glucose **rises**.
- Result: blood glucose stays around 70–110 mg per 100 mL (Campbell p. 893).



Blood sugar control: insulin lowers blood glucose after eating and glucagon raises it between meals, keeping it near 70–110 mg per 100 mL.

Source: Campbell Biology, Figure 41.23, p. 928

5. Example: water balance (M&L pp. 24, 932)

- **Sweating** makes you lose water, so dissolved materials in the blood become more concentrated → the hypothalamus detects this → it triggers ADH (kidneys save water) and **thirst** (you drink) → water level **rises** back to normal.
- **After drinking** a lot, less ADH is released → the kidneys remove more water → the level **falls** back.

6. Why it's a *dynamic equilibrium*. Responses aren't instant, so conditions wobble a little above and below the set point before being corrected. Homeostasis "moderates but doesn't eliminate changes" (Campbell p. 894). Stability comes from *constant correction*, like a driver steering to stay in the lane (M&L p. 907).

7. What about positive feedback? Positive feedback *amplifies* a change instead of reversing it, so it does **not** maintain homeostasis. In the body it's used only for processes that need to *finish*: childbirth contractions and blood clotting (Campbell pp. 9, 894). In natural systems it's often harmful, because it pushes things to extremes (teacher's slide 3). The climate examples from class:

Slide	Loop	Type	Why
6	More CO ₂ → warmer → more photosynthesis → CO ₂ removed → cooling	Negative	The result opposes the warming.
7	Warmer → ice melts → lower albedo → more heat absorbed → warmer	Positive	The result adds to the warming.
8	Warmer → permafrost melts → methane released → warmer	Positive	The result adds to the warming.
9	Warmer → more evaporation → more clouds → more reflection → cooler	Negative	The result opposes the warming.

Slide	Loop	Type	Why
10	Warmer → faster decomposition → more CO ₂ → warmer	Positive	The result adds to the warming.

How to tell them apart: follow the loop all the way around. If you end up pushing the original condition *back*, it's negative. If you push it *further the same way*, it's positive.

8. When feedback fails. “Any breakdown of homeostasis may have serious or even fatal consequences” (M&L p. 25). If sweating can't keep up on a very hot day, body temperature climbs out of range (heat-stroke). If the insulin loop doesn't work, blood glucose stays high (diabetes).

Tips for the Feedback practice worksheet

- For **negative feedback**, put a *measurable* condition in the center box and draw two loops, one for “too high” and one for “too low.” Everyday examples: water temperature in a shower (too hot → turn the cold up; too cold → turn the hot up), or hunger level (hungry → eat → full → stop eating).
- For **positive feedback**, make sure every arrow makes the condition *more extreme*. Everyday examples: a microphone near a speaker (sound → amplified → picked up again → louder squeal), or a rumor spreading (more people know → more people tell others → even more people know).
- Don't use technical examples you found online, such as blood glucose. The worksheet asks for your own everyday examples.

Sources

- Miller, K. R., & Levine, J. S. *Miller & Levine Biology*. Pearson, 2019. Lesson 1.1 (p. 12), Lesson 1.2 (pp. 15, 21), Lesson 1.3 (pp. 22–25), Lesson 27.1 (pp. 907–908), Lesson 27.3 (p. 932).
- Urry, L. A., et al. *Campbell Biology*. Ch. 1, Concept 1.1 (pp. 3, 9); Ch. 8, Concept 8.1 (p. 146); Ch. 40, Concept 40.2 (pp. 893–894).
- Teacher materials: *Ecology Unit Guide* (KJ), *What Is Life? Skeleton Notes*, *Feedback Mechanisms and Homeostasis* slides, *Feedback Mechanisms and Homeostasis Practice*.